Jasper Grant

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Education

MASc. in Electrical Engineering, GPA: 4.23/4.30 – Dalhousie University May 2026
Relevant Coursework: Autonomous Robotic Navigation in Harsh Environments, Applied ML for Software Engineers
BEng. in Electrical Engineering, GPA: 3.77/4.30 – Dalhousie University Dec 2024

Experience

Mars Rover Autonomy Intern, NASA Jet Propulsion Lab (JPL) – Pasadena, CA, USA Jan 2027 – Apr 2027
- Will develop visual navigation capabilities to advance Mars rover autonomy for future mission operations.

Lunar Rover Research Intern, NASA Ames Research Center – Mountain View, CA, USA Jun 2026 – Dec 2026
- Developed TIDE, a dynamics-based terrain identification method, validated on NASA's KREX2 rover.
- Generated a visual-inertial-LiDAR lunar rover traverse dataset in a photorealistic Gazebo-based simulator to support training and evaluation of next-generation lunar navigation algorithms.

MITACS Intern, MDA Space: Robotics and Space Operations – Brampton, ON, Canada Jan 2026 – Apr 2026
- Implemented abstract interfaces in MDA AutoNav (autonomous rover navigation system) to decouple core logic from ROS, improving modularity and portability.
- Implemented a deformable terrain testbed using Project Chrono to evaluate software traction control methods.

Teaching Assistant, Dalhousie University – Halifax, NS, Canada May 2025 – Dec 2025
- Supported student robotics development in ROS2 across two upper-year courses (ECED3901, MECH4640) by configuring environments, troubleshooting, and developing technical examples and documentation.

Software Engineering Co-op Student, MDA Space: Geointelligence – Halifax, NS, Canada May 2023 – Dec 2023
- Developed and maintained features in a large microservices architecture project using Angular, Go, and Java Spring, improving functionality and resolving defects across front-end and back-end systems.
- Automated import of Katalon Studio test results into Jira, reducing manual QA effort; adopted across multiple MDA projects. Also built regression tests in Groovy and Cucumber.

Software Development Co-op Student, ISE – Port Coquitlam, BC, Canada Sep 2022 – Dec 2022
- Developed a Qt-based UI for AUV mission planning, enabling operators to define and execute lawnmower survey patterns over irregular survey regions.
- Implemented a C++ NMEA string interpreter to parse GPS communications and extract positional measurements for integration into onboard navigation systems.

Student Research Assistant, Dalhousie Intelligent Systems Lab – Halifax, NS, Canada Apr 2021 – Aug 2021
- Designed PCBs in Eagle for power distribution and solenoid control in a robotic spacecraft simulator.
- Developed C++ and Python implementations for ROS-based robot communication in the Gazebo physics simulator, supporting validation of multi-robot interactions.

Projects and Publications

TIDE: Terrain Identification via Dynamic Effects in a Parametric Motion Model IRG Link
- Dynamics-based terrain identification on NASA's KREX2 rover. Submitted to ICRA 2027.

Terrain-Adaptive Strategies to Prevent and Recover from Rover Wheel-Slip IEEE Xplore Link
- Developed terrain-adaptive lunar rover control using MAP-Elites. Presented at IEEE Aerospace 2026.

3DOF Proximity Flight Emulation with a Planar Air-Bearing Testbed IEEE Xplore Link
- Constructed a Control Moment Gyroscope (CMG) for a planar air-bearing testbed. Presented at ICACR 2024.

Mine Countermeasures (MCM) – Autonomy Infrastructure Development DRDC link
- Evaluated behavior arbitration strategies for autonomous underwater vehicle (AUV) autonomy in support of a Defence Research and Development Canada (DRDC) contractor report.

Skills

Robotics: ROS, ROS2, Gazebo, Project Chrono, Simulink
Software: C++, Python, C, Go, TypeScript, Groovy
CAD: Fusion 360, Eagle PCB, SolidWorks